



JAIN
DEEMED-TO-BE UNIVERSITY

SCHOOL OF
COMPUTER
SCIENCE AND IT

Department of CS & IT
Programme: MCA AI & ML

ACTIVITY - 3: Mini Project
Predictive Analysis and Data Analysis
23MCAAI303

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Certificate

This is to certify that Prangya Kumari Nayak and K. Videth satisfactorily completed the course of Activity - 3: Mini Project prescribed by Jain (Deemed-to-be University) for the semester III MCA degree course in the year 2024-2026.

Date: 3rd October 2025

Prangya Kumari Nayak



Signature of Student

Introduction

Customer churn is one of the most pressing challenges for companies in the telecommunications industry. In competitive markets, customers can switch providers easily if they find better pricing, service quality, or features. This makes churn not just a metric, but a critical indicator of long-term profitability. According to industry studies, acquiring a new customer can cost up to five times more than retaining an existing one. Therefore, reducing churn has a direct impact on financial performance.

In this project, we analyze customer churn using the **Telco Customer Churn Dataset**, widely used in research and case studies. Our analysis combines statistical exploration, visualization, and dashboarding to understand:

- **Patterns of churn across demographics.**
- **Service features that drive customer decisions.**
- **The role of contracts, payments, and tenure.**
- **Strategies companies can adopt to mitigate churn.**

This analysis is presented using two main platforms: **Google Colab** for data preprocessing and visualization, and **Streamlit** for building interactive dashboards.

Dataset Description

The dataset consists of **7043 customer records** with **21 attributes**, capturing demographic details, services subscribed, billing methods, and churn outcomes.

1. Key Features

Feature	Description
customerID	Unique ID for each customer
gender	Male or Female
SeniorCitizen	1 if senior citizen, 0 otherwise
Partner	Whether the customer has a partner
Dependents	Whether the customer has dependents
tenure	Number of months the customer has been with the company
PhoneService	Whether the customer has phone service
InternetService	Type of internet service (DSL, Fiber, None)
Contract	Contract type (Month-to-month, One year, Two year)
PaymentMethod	Payment method used
MonthlyCharges	Customer's monthly charges
TotalCharges	Total amount charged to the customer
Churn	Whether the customer left within the last month (Yes/No)

2. Initial Observations

- The dataset covers a wide variety of customer types.
- Some variables (like **TotalCharges**) have missing or inconsistent values.
- Several categorical columns need encoding before modeling.
- The dataset is imbalanced: churn rate is around **26.5%**, meaning most customers stay but a significant fraction leaves.

Data Cleaning & Preprocessing

Data preprocessing ensures accuracy and reliability. Steps included:

1. **Missing Values:** Handled empty or whitespace records in **TotalCharges**. These were converted to numeric and imputed where possible.
2. **Data Types:** Converted columns like tenure and charges into appropriate numeric types.
3. **Categorical Encoding:** Applied one-hot encoding for categorical features (e.g., Contract, InternetService).
4. **Normalization:** Standardized continuous variables (MonthlyCharges, TotalCharges) for visualization and later use in modeling.
5. **Class Balance:** Checked churn class distribution. Though slightly imbalanced, no resampling was applied at this stage since analysis was descriptive.

Exploratory Data Analysis (EDA)

1. Customer Demographics

- **Gender:** Churn rates between men and women were nearly identical, indicating gender is not a strong predictor.
- **Senior Citizens:** Higher churn compared to younger customers.
- **Dependents:** Customers without dependents churned more often.

2. Service Features

- **Internet Service:** Fiber optic customers had the highest churn rates. DSL users were more loyal.

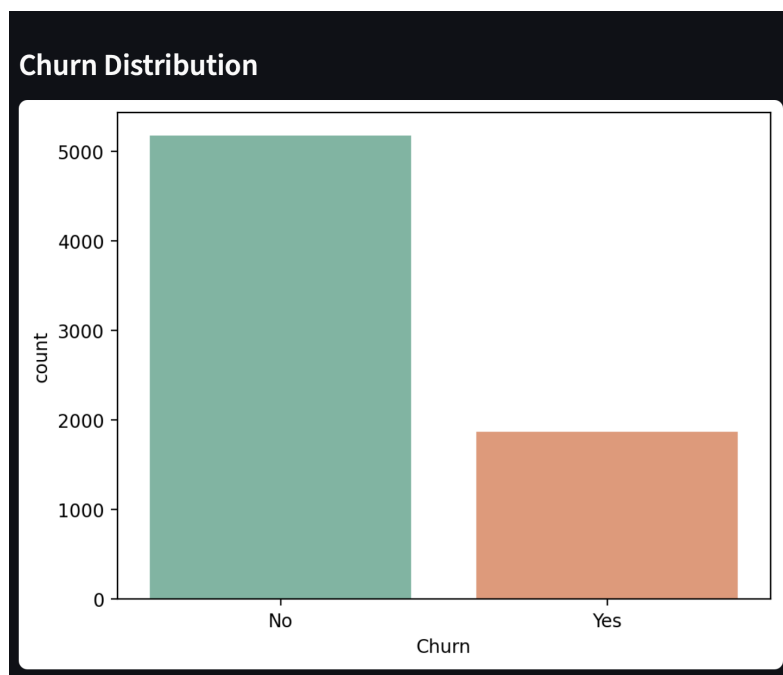
- **Contract:** Month-to-month contracts were strongly associated with churn. Long-term contracts retained customers better.
- **Payment Methods:** Customers using electronic checks showed higher churn, possibly due to billing friction or dissatisfaction.

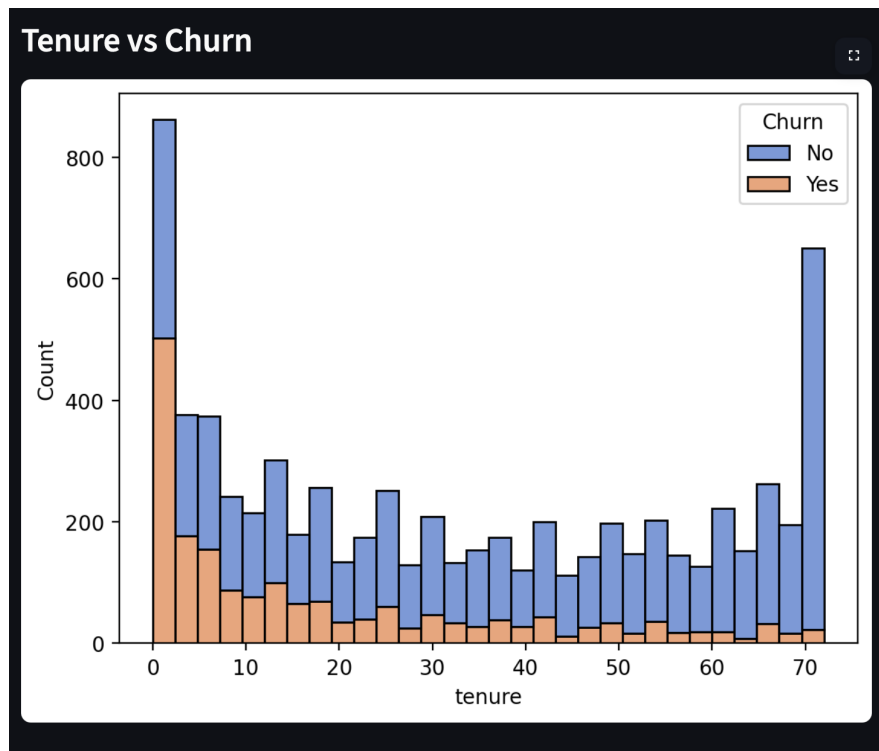
3. Financial Features

- **Monthly Charges:** Customers paying higher monthly fees churned more.
- **Tenure:** Customers with short tenure (0–12 months) had the highest churn. Retention improves after the first year.

4. Visualizations

- **Bar Charts:** Showed categorical distributions.
- **Histograms:** Revealed skewness in tenure and charges.
- **Heatmaps:** Displayed correlation between churn and features like contract type, charges, and tenure.





Dashboard Overview

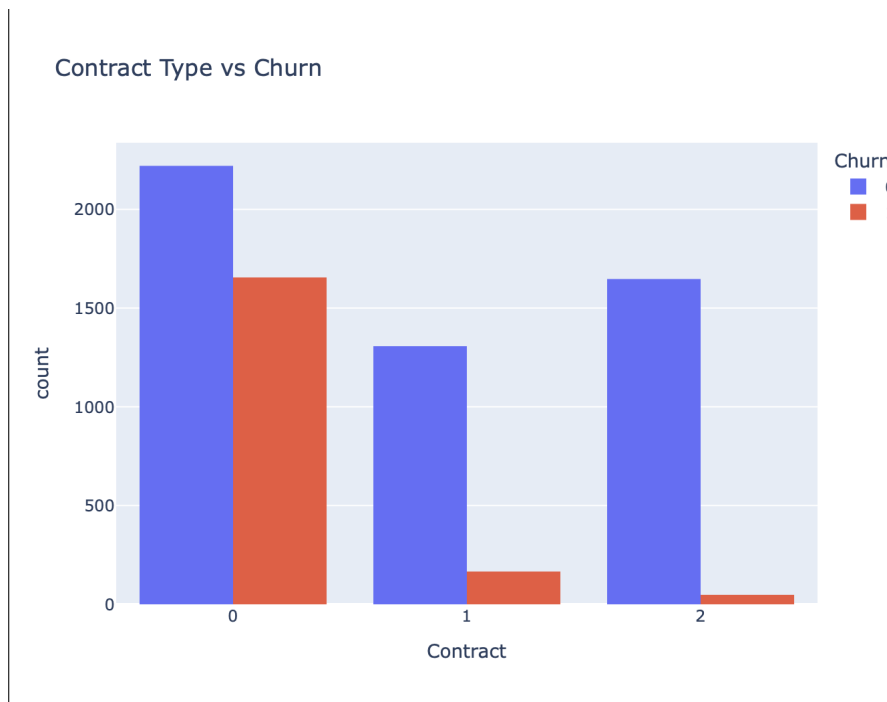
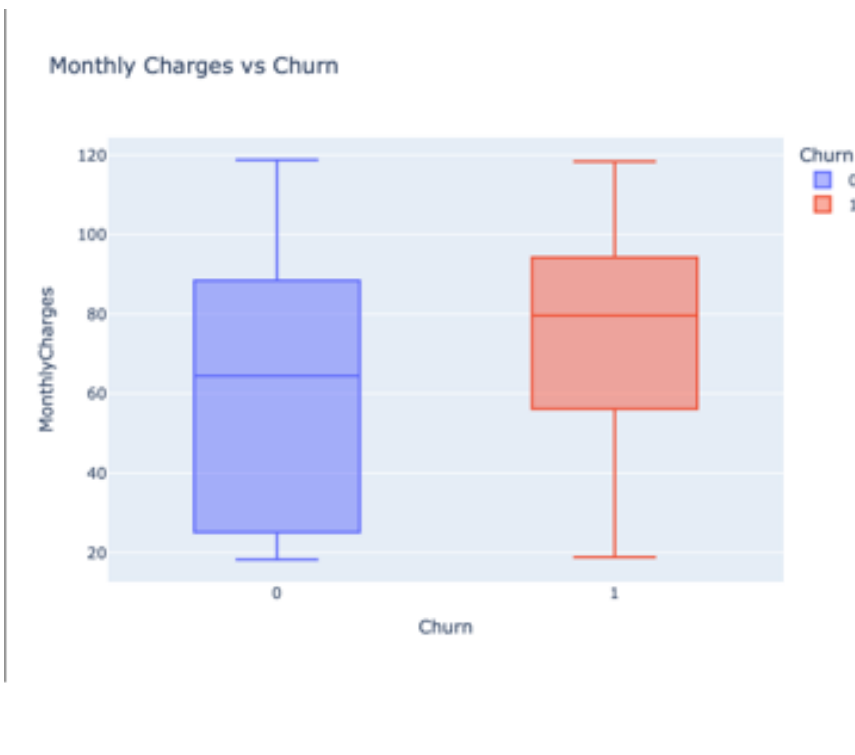
1. Google Colab Dashboard

- Built using **Matplotlib and Seaborn**.
- Included interactive plots with filtering options.
- Provided correlation matrices and summary statistics.

2. Streamlit Dashboard

- Allowed users to filter churn data by **gender, contract type, and internet service**.
- Updated charts dynamically based on selections.

- Gave decision-makers a hands-on tool to explore churn behavior.



Key Insights

1. **Contract Type is critical:** Month-to-month customers churn at more than double the rate of yearly contract customers.
 2. **Service Bundling helps:** Customers with multiple services (phone + internet + TV) churn less.
 3. **High Charges drive churn:** Customers with bills above \$80/month were significantly more likely to leave.
 4. **Early-stage churn is common:** Customers often leave within their first year. Retention programs should focus on onboarding.
 5. **Electronic Check customers are at risk:** They showed the highest churn, perhaps due to outdated or inconvenient billing practices.
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Recommendations

1. **Incentivize long-term contracts:** Discounts or loyalty points for yearly contracts can reduce churn.
2. **Target early-tenure customers:** Personalized offers in the first 6 months may prevent early dropouts.
3. **Review pricing for high-charge customers:** Offer flexible plans or discounts to prevent dissatisfaction.
4. **Encourage auto-pay methods:** Promote credit card or bank transfer payments instead of electronic checks.
5. **Expand DSL/fiber quality:** Since fiber optic customers churn more, improving service quality can help.

Limitations

- Dataset lacks behavioral features like call logs, data usage, or customer support interactions.
- Churn reasons are inferred indirectly from features, not customer feedback.
- Dataset is a snapshot and may not reflect long-term market trends.

Future Scope

1. **Predictive Modeling:** Use machine learning algorithms (Logistic Regression, Random Forest, XGBoost) to predict churn.
2. **Customer Segmentation:** Apply clustering to identify distinct customer groups.
3. **Sentiment Analysis:** Incorporate customer complaints and feedback for richer churn analysis.
4. **Deployment:** Host the Streamlit app online for company-wide access.
5. **Real-Time Dashboards:** Integrate with live databases to track churn continuously.

Conclusion

Customer churn remains a challenge for telecom companies, but with proper analysis, patterns can be identified and acted upon. This project demonstrates how demographic, service, and billing data provide meaningful insights into customer

behavior. By adopting the suggested recommendations and expanding into predictive modeling, telecom providers can reduce churn and improve customer satisfaction.

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Datasets & Tools

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